

Bluestock Mutual Fund Analytics Capstone

End-to-End Data Engineering, Risk Modeling, & BI Dashboarding

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The Data Fragmentation Problem

The mutual fund industry generates massive data streams, but they are rarely unified.

- **Daily NAVs:** Sit in isolated pricing feeds.
- **Monthly AUMs:** Locked in distinct compliance disclosures.
- **Transactions:** Buried in granular investor ledgers.

"Without a central key connecting behavior to performance, AMCs miss critical signals like at-risk capital and hidden concentration risks."



Core Project Objectives



Data Engineering

Design a robust ETL pipeline in Python to ingest, clean, and centralize raw CSV feeds into a unified SQLite star-schema database.



Quantitative Performance

Compute standardized financial metrics (Alpha, Beta, CAGR, and Sharpe Ratio) to build an objective Top 40 Fund Scorecard.



Advanced Risk Analytics

Move beyond basic volatility by modeling extreme tail risk (Historical VaR), portfolio concentration (HHI), and investor SIP continuity.

ETL & Analytics Architecture



1. Ingestion & Cleaning

Pandas/NumPy utilized to forward-fill missing NAVs, enforce ISO date formats, and cap severe transaction outliers.



2. Centralized Storage

Data structured into a relational SQLite database (`bluestock_mf.db`) using a star-schema (Facts linked to `Dim_Fund`).



3. Analytics Engine

Python modules execute CAPM formulas and historical simulations, outputting the master `fund_scorecard.csv`.



4. BI Visualization

Power BI connects to the processed datasets to deliver a 4-page interactive dashboard for non-technical stakeholders.

Data Processing Scale (2022-2025)

104K+

DAILY NAV RECORDS

51K+

CLEANED TRANSACTIONS

< 0.5%

TOTAL DATA LOSS

Rigorous programmatic cleaning ensured high relational integrity before analytics commenced.

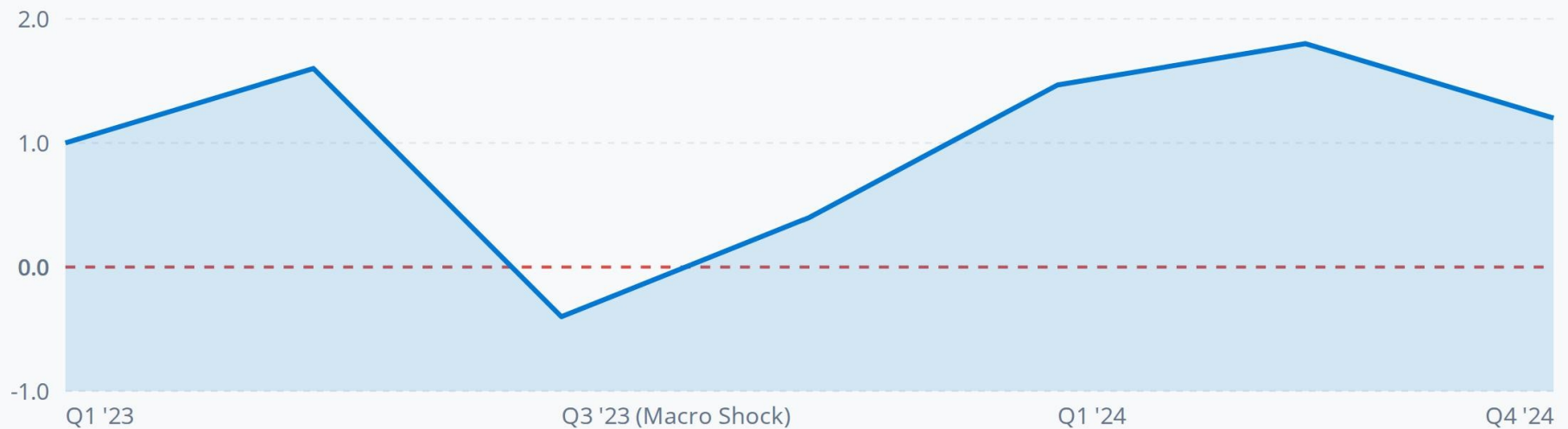
Fund Scorecard: Top vs Bottom Performers

Analysis of the Top 40 funds revealed that Large Cap funds offered the most stable risk-adjusted returns.

AMFI CODE	CATEGORY	ALPHA (EXCESS)	BETA (VOL.)	CAGR	SHARPE RATIO
118632	Large Cap	0.04	0.85	18.2%	2.10 (Leader)
119092	Multi Cap	0.02	0.92	15.4%	1.50
120503	Mid Cap	0.01	1.05	14.1%	1.20
119551	Small Cap	-0.01	1.18	12.8%	0.80
120841	Sectoral	-0.03	1.25	9.5%	0.60 (Laggard)

Dynamic Volatility: 90-Day Rolling Sharpe

Point-in-time metrics are deceptive. Rolling analytics prove that risk-adjusted returns plunge below zero during macroeconomic shocks (e.g., rate hikes) before aggressively recovering.



Extreme Tail Risk: 95% Historical VaR

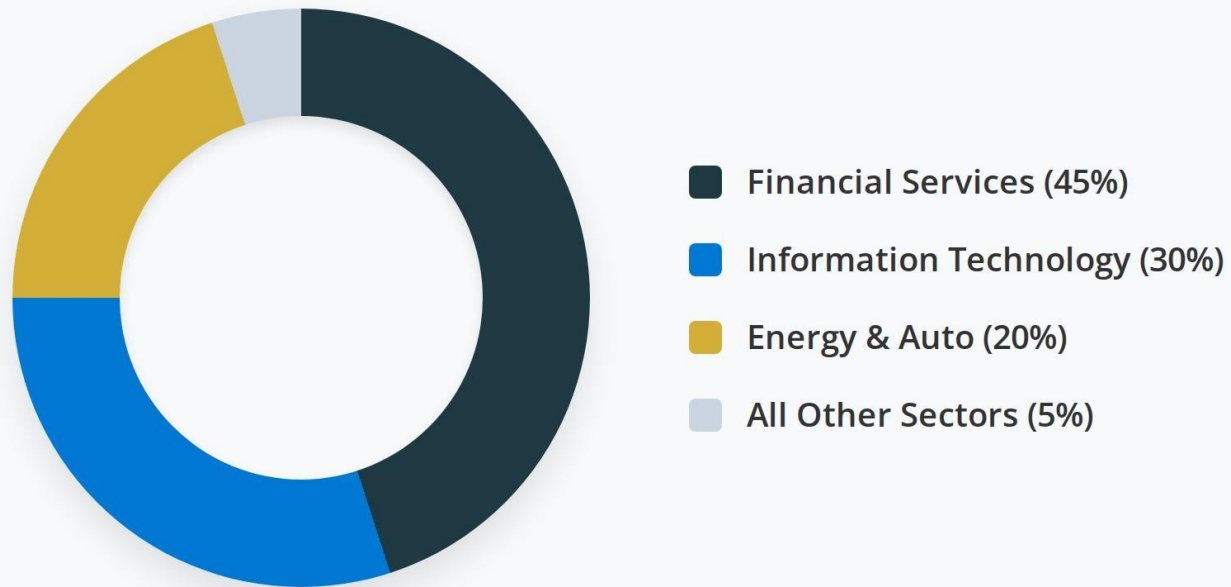
Value at Risk (VaR) estimates the maximum expected 1-day loss. Analysis shows that the highest-risk funds failed to deliver proportional returns, suffering from severe volatility drag.



**A 95% VaR of -2.15% implies a 5% historical probability of losing more than 2.15% of capital in a single trading day.*

Covert Concentration Risk (HHI)

Applying the Herfindahl-Hirschman Index (HHI) to sector weights revealed that several "Diversified" funds hold covert overweight positions, scoring near 2,500 (highly concentrated).



Investor Behavior: SIP Continuity

SIPs form the backbone of AMC revenue, but continuity analysis isolated a critical vulnerability.

- **SIP Dominance:** Systematic Investment Plans account for 83.5% of all tracked transaction volume.
- **The Vulnerability:** 10-15% of seasoned investors (6+ previous SIPs) fall into the "at-risk" category.
- **The Metric:** "At-risk" is defined by exhibiting a gap of more than 35 days between investments.
- **The Cause:** Often correlates with minor macro-corrections or failed bank auto-debit setups.

 Dashboard data visualization showing transaction trends

Strategic Recommendations for AMCs



Deploy automated trigger workflows (SMS/Email) specifically targeting the "at-risk" investor cohort flagged by the 35-day continuity algorithm to recover failed auto-debits.



Shift away from static risk labels (Low/Med/High). Utilize rolling-risk metrics like the 90-Day Rolling Sharpe and Historical VaR to better manage investor expectations during shocks.



Integrate internal HHI scoring alerts to ensure funds marketed and sold as "Diversified Equity" do not silently breach safe concentration thresholds in IT and Financials.

Questions?

Thank you for your time.

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 github.com/adityaraj9798/Mutual-Fund-Analytics

Image Sources



https://static.vecteezy.com/system/resources/previews/075/683/084/non_2x/abstract-financial-growth-background-with-glowing-purple-data-chart-and-rising-trend-line-modern-business-analytics-concept-design-with-copy-space-for-presentation-report-or-marketing-use-vector.jpg

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